

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets

QUESTIONS: 05

Total Marks:50

Code of Conduct:

I K. Venkata Karthik / 192372092 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Venkata Karthik

Q1. Dataset: Hospital Patient Records

PID Symptoms Observed

P1 Fever, Cough

P2 Fever, Headache

P3 Cough, Headache

P4 Fever, Cough, Headache

P5 Fever, Fatigue

Question:

Apply association rule mining with minimum support = 50% and confidence = 70%, under the constraint that all generated rules must include "Fever". Identify the valid association rules and justify why they satisfy the given constraints.

Answer:

Given Dataset

There are 5 patients, so:

$$\text{Minimum support count} = 50\% \times 5 = 2.5$$

Therefore, an itemset must occur in at least 3 patients.

Patient	Symptoms
P1	Fever, Cough

P2	Fever, Headache
P3	Cough, Headache
P4	Fever, Cough, Headache
P5	Fever, Fatigue

Step 1: Support of individual symptoms

- **Fever** $\rightarrow$ P1, P2, P4, P5 = $4/5 = 80\%$
- **Cough** $\rightarrow$ P1, P3, P4 = $3/5 = 60\%$
- **Headache** $\rightarrow$ P2, P3, P4 = $3/5 = 60\%$
- **Fatigue** $\rightarrow$ P5 = $1/5 = 20\%$

So the frequent symptoms are Fever, Cough, and Headache.

Step 2: Frequent itemsets containing Fever

{Fever, Cough}

Occurs in P1 and P4:

$$Support = \frac{2}{5} \times 100 = 40\%$$

Since $40\% < 50\%$, it is **not valid**.

{Fever, Headache}

Occurs in P2 and P4:

$$Support = \frac{2}{5} \times 100 = 40\%$$

Since $40\% < 50\%$, it is **not valid**.

{Fever, Cough, Headache}

Occurs only in P4:

$$Support = \frac{1}{5} \times 100 = 20\%$$

So it is also **not valid**.

Step 3: Association rules

Because every rule must contain **Fever**, possible rules include:

- Fever $\rightarrow$ Cough
- Fever $\rightarrow$ Headache
- Cough $\rightarrow$ Fever
- Headache $\rightarrow$ Fever

- Fever, Cough → Headache
- Fever, Headache → Cough

Now check confidence:

1. Fever → Cough

$$\text{Confidence} = \frac{\text{Support}(F, \text{everCough})}{\text{Support}(\text{Fever})} = \frac{2/5}{4/5} = 50\%$$

50% < 70% → Invalid

2. Fever → Headache

$$\text{Confidence} = \frac{2/5}{4/5} = 50\%$$

50% < 70% → Invalid

3. Cough → Fever

$$\text{Confidence} = \frac{2/5}{3/5} = 66.67\%$$

66.67% < 70% → Invalid

4. Headache → Fever

$$\text{Confidence} = \frac{2/5}{3/5} = 66.67\%$$

66.67% < 70% → Invalid

Final Answer

There are no valid association rules satisfying both:

- Minimum support = **50%**
- Minimum confidence = **70%**
- Every rule must include **Fever**

Although **Fever** itself has sufficient support (**80%**), every symptom combination containing Fever has support below 50%, and the possible Fever-related rules also fail the 70% confidence requirement.

Therefore, the set of valid association rules is: *None*.

Q2. Dataset: Online Learning Platform

Student ID Courses Enrolled

S1 Python, SQL

S2 Python, Data Science

S3 SQL, Machine Learning

S4 Python, SQL, Data Science

S5 Python, Machine Learning

Question:

Using the dataset, generate frequent itemsets with minimum support = 40%, applying the constraint that each itemset must contain at least two courses. Explain how the constraint helps in pruning candidate itemsets during mining.

Answer:

Given Dataset:

Student ID	Courses Enrolled
S1	Python, SQL
S2	Python, Data Science
S3	SQL, Machine Learning
S4	Python, SQL, Data Science
S5	Python, Machine Learning

Minimum Support = 40%

Number of students = 5

Minimum support count:

$$40\% \times 5 = 2$$

Therefore, an itemset must occur in **at least 2 students** to be considered frequent.

Step 1: Generate 1-Itemsets

Itemset	Support Count	Support
{Python}	4	80%
{SQL}	3	60%
{Data Science}	2	40%
{Machine Learning}	2	40%

All four courses satisfy the minimum support. However, the given constraint requires at least two courses in each itemset, so these 1-itemsets are not considered in the final result.

Step 2: Generate 2-Itemsets

Itemset	Support Count	Support	Result
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{Python, SQL}	2	40%	Frequent
{Python, Data Science}	2	40%	Frequent
{Python, Machine Learning}	1	20%	Not Frequent
{SQL, Data Science}	1	20%	Not Frequent
{SQL, Machine Learning}	1	20%	Not Frequent
{Data Science, Machine Learning}	0	0%	Not Frequent

Step 3: Generate 3-Itemsets

The possible 3-itemsets are:

- {Python, SQL, Data Science} → Support = $1/5 = 20\%$
- {Python, SQL, Machine Learning} → Support = 0%
- {Python, Data Science, Machine Learning} → Support = 0%
- {SQL, Data Science, Machine Learning} → Support = 0%

Since all these itemsets have support less than 40%, no 3-itemset is frequent.

Final Frequent Itemsets

The frequent itemsets satisfying both the minimum support of 40% and the constraint of at least two courses are:

1. {Python, SQL} – Support = 40%
2. {Python, Data Science} – Support = 40%

Effect of the Constraint on Pruning

The constraint “**each itemset must contain at least two courses**” helps to prune the search space by eliminating all single-course itemsets from consideration in the final result. It focuses the mining process only on combinations containing two or more courses. This reduces the number of candidate itemsets and makes the mining process more efficient.

Conclusion: The valid frequent itemsets are {Python, SQL} and {Python, Data Science}, both having 40% support.

Q3. Dataset: Library Borrowing Records

Transaction ID Books Borrowed

T1 Data Mining, Python Programming

T2 Data Mining, Database Systems

T3 Python Programming, Database Systems

T4 Data Mining, Python Programming, Database Systems

T5 Data Mining, Artificial Intelligence

Question:

Apply the Apriori algorithm with the constraint that the maximum itemset size is 3. Identify the frequent itemsets and explain how the constraint reduces computational complexity during candidate generation.

Answer:

Given Dataset

Transaction ID	Books Borrowed
T1	Data Mining, Python Programming
T2	Data Mining, Database Systems
T3	Python Programming, Database Systems
T4	Data Mining, Python Programming, Database Systems
T5	Data Mining, Artificial Intelligence

Assume the minimum support is 40% (as in the previous question).

Number of transactions = 5

$$\text{Minimum Support Count} = 40\% \times 5 = 2$$

Therefore, an itemset must occur in at least 2 transactions to be frequent.

Step 1: Generate 1-Itemsets

Itemset	Support Count	Support
{Data Mining}	4	80%
{Python Programming}	3	60%
{Database Systems}	3	60%
{Artificial Intelligence}	1	20%

Frequent 1-itemsets are:

- {Data Mining}
- {Python Programming}
- {Database Systems}

{Artificial Intelligence} is removed because its support is only 20%.

Step 2: Generate 2-Itemsets

Using the frequent 1-itemsets, the following candidates are generated:

Itemset	Support Count	Support	Result
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{Data Mining, Python Programming}	2	40%	Frequent
{Data Mining, Database Systems}	2	40%	Frequent
{Python Programming, Database Systems}	2	40%	Frequent

Therefore, all three 2-itemsets are frequent.

Step 3: Generate 3-Itemsets

Since the maximum itemset size is 3, we can generate 3-itemsets but cannot generate any 4-itemsets.

The only possible 3-itemset is:

{Data Mining, Python Programming, Database Systems}

It occurs only in T4.

$$Support = \frac{1}{5} \times 100 = 20\%$$

Since $20\% < 40\%$, it is not frequent.

Final Frequent Itemsets

Itemset	Support
{Data Mining}	80%
{Python Programming}	60%
{Database Systems}	60%
{Data Mining, Python Programming}	40%
{Data Mining, Database Systems}	40%
{Python Programming, Database Systems}	40%

There are no frequent 3-itemsets.

Effect of Maximum Itemset Size Constraint

The constraint maximum itemset size = 3 limits the Apriori algorithm to generating itemsets containing at most three books. Therefore, after generating 3-itemsets, the algorithm stops and does not generate 4-itemsets or larger itemsets.

Without this constraint, Apriori would continue generating larger candidate itemsets, increasing the number of candidates, database scans, memory usage, and computation time. Thus, the constraint reduces the search space and computational complexity of candidate generation.

Conclusion

Using Apriori with a minimum support of 40% and maximum itemset size of 3, the frequent itemsets are {Data Mining}, {Python Programming}, {Database Systems}, {Data Mining,

Python Programming}, {Data Mining, Database Systems}, and {Python Programming, Database Systems}.

Q4. Dataset: University Course Hierarchy

Student ID Courses Taken

S1 Engineering, Computer Science, Machine Learning

S2 Engineering, Information Technology

S3 Engineering, Computer Science

S4 Engineering, Computer Science, Machine Learning

S5 Engineering, Information Technology

Question:

Apply multilevel association rule mining with the constraint that rules should be generated only at the "Computer Science" level. Explain how the course hierarchy affects the discovery of association rules.

Answer:

Given Dataset

Student	Courses Taken
S1	Engineering, Computer Science, Machine Learning
S2	Engineering, Information Technology
S3	Engineering, Computer Science
S4	Engineering, Computer Science, Machine Learning
S5	Engineering, Information Technology

Constraint: Generate rules only at the Computer Science level.

Analysis

At the Computer Science level:

- Computer Science appears in S1, S3, and S4 $\rightarrow 3/5 = 60\%$
- Machine Learning appears in S1 and S4 $\rightarrow 2/5 = 40\%$
- Computer Science + Machine Learning appears in S1 and S4 $\rightarrow 2/5 = 40\%$

Therefore, the association rule is:

Computer Science $\rightarrow$ Machine Learning

Support:

$$\frac{2}{5} \times 100 = 40\%$$

Confidence:

$$\frac{Support(C, SML)}{Support(CS)} = \frac{2}{3} \times 100 = 66.67\%$$

Effect of Course Hierarchy

The hierarchy allows courses to be analyzed at different levels, such as Engineering → Computer Science → Machine Learning. Since the constraint specifies the Computer Science level, rules involving higher-level categories such as Engineering are excluded. This reduces the search space and focuses the mining process on more specific and meaningful relationships.

Final Rule: Computer Science → Machine Learning (Support = 40%, Confidence = 66.67%).

Q5. Dataset: Insurance Customer Records

CID Age Group Income Level Policy Purchased

C1 Young High Health Insurance

C2 Middle Medium Vehicle Insurance

C3 Young Low Travel Insurance

C4 Senior High Health Insurance

C5 Young Medium Vehicle Insurance

Question:

Apply multidimensional association rule mining with the constraint Age Group = "Young". Identify meaningful patterns involving income level and policy type, and explain how these patterns can support targeted marketing strategies.

Answer:

Given Dataset

CID	Age Group	Income Level	Policy Purchased
C1	Young	High	Health Insurance
C2	Middle	Medium	Vehicle Insurance
C3	Young	Low	Travel Insurance
C4	Senior	High	Health Insurance
C5	Young	Medium	Vehicle Insurance

Constraint: Age Group = Young

Young customers are C1, C3, and C5.

Meaningful Patterns

Income Level	Policy	Customers
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High	Health Insurance	1
Low	Travel Insurance	1
Medium	Vehicle Insurance	1

The patterns show that among young customers:

- High-income → Health Insurance
- Medium-income → Vehicle Insurance
- Low-income → Travel Insurance

Marketing Strategy

These patterns can help the insurance company create **targeted offers** for young customers. High-income young customers can be offered premium health plans, medium-income customers can receive vehicle insurance offers, and low-income customers can be targeted with affordable travel insurance plans.

Conclusion: Restricting the analysis to Young customers makes the patterns more specific and helps the company design suitable insurance products for different income groups.